StockEvent Lens — Batch 022 Intelligence Dashboard

Articles 106–110 Analysis, Vector Embedding Sensitivity & Conflict Thresholds

Batch ID
`batch_22 (Articles 106-110)`

TEST 106
Original 100% Base
NVIDIA CEO Jensen Huang expects chip sales volume to double (2x) next year[cite: 24, 28].

TEST 107
Repeat 96% Overlap
Restates Huang's 2x chip sales growth forecast at UK summit without new facts[cite: 25, 28].

TEST 108
Update +25% New Info
Samsung (+3.37%) and SK hynix (+6.42%) rally on accelerated HBM4/HBM4E memory shift[cite: 26, 28].

TEST 109
Conflict Dispute (3x)
Contradicts sales guidance: claims Huang expects chip sales volume to triple (3x)[cite: 27, 28].

TEST 110
Different 0% Overlap
NVIDIA Vera Rubin NVL72 MLPerf Inference v6.1 benchmark debut starting Group 044[cite: 23, 28].

Technical Analysis: Dense Vector Blindness vs. Temporal Contradiction Metrics

A core challenge in automated financial news aggregation is that dense vector models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) compute cosine similarity based on *semantic context* rather than *executive growth multiplier accuracy*. Modifying volume targets (e.g., "twice as many" [2x] in test_106 vs "three times as many" [3x] in test_109) yields high vector similarity (~0.93) because both texts share identical surrounding context tokens ("Jensen Huang", "Nvidia", "King Charles III", "summit in Scotland", "chip sales next year").

```
// Formula: Formal Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a hard numerical multiplier or growth target dispute.
```

1. Dense Embedding Blindness
Dense embeddings project words into high-dimensional space where "twice as many chips" and "three times as many chips" share near-identical context embeddings. Thus, pure vector cosine distance fails to flag this quantitative conflict.

2. LLM Evaluation Override
To prevent corrupting financial guidance and market demand forecasts, the LLM reasoning layer performs explicit attribute-value multiplier matching, overriding high vector similarity when $\Delta F_{contradiction} = 1.0$ is detected.

Article Relationship & Percentage Breakdown (Batch 022)

FILE IDENTIFIER	PATTERN LABEL	KEYWORDS IDENTIFIED	SIMILARITY %	DELTA %	CLASSIFICATION RATIONALE
test_106_original	Original (%5=1)	Jensen Huang, double chip sales (2x), King Charles III summit, AI safety[cite: 24, 28]	100%	Baseline	Establishes breaking news frame on Huang foreseeing 2x chip volume growth next year[cite: 24, 28].
test_107_repeat	Repeat (%5=2)	Nvidia double sales next year, Scotland AI summit, King Charles III[cite: 25, 28]	96%	0% New Facts	Restates Huang's 2x chip sales doubling statement and UK summit context without new developments[cite: 25, 28].
test_108_update	Update (%5=3)	**Samsung Electronics +3.37%**, SK hynix +6.42%, HBM4 & HBM4E acceleration[cite: 26, 28]	82%	+25% Market Impact	Advances event state to Korean chip stock rally (Samsung, SK hynix) and HBM4/HBM4E memory acceleration[cite: 26, 28].
test_109_conflict	Conflict (%5=4)	**Triple chip sales (3x) dispute**, three times as many chips claim[cite: 27, 28]	93% Match	$\Delta F_{contradiction} = 1.0$	Directly disputes sales forecast, claiming Huang expects chip sales volume to triple (3x) rather than double (2x)[cite: 27, 28].
test_110_different	Different (%5=0)	NVIDIA Vera Rubin NVL72, MLPerf Inference v6.1, Qwen3-VL, DeepSeek-R1[cite: 23, 28]	0%	Group 044 Trigger	Zero domain overlap; triggers standalone Group 044 event cluster for Vera Rubin NVL72 MLPerf v6.1 results[cite: 23, 28].

Group 043 — Consolidated Event Summary

4 Articles (106–109)

NVIDIA CEO Jensen Huang stated at a U.K. summit in Scotland with King Charles III that NVIDIA expects to sell twice as many chips next year as this year due to surging global AI demand (Articles 106–107)[cite: 24, 25, 28]. Global supply chain reaction advanced as **Korean chipmakers Samsung Electronics (+3.37%) and SK hynix (+6.42%) rallied strongly on expectations that 2x GPU volume growth will accelerate the transition to HBM4 and HBM4E memory** (Article 108)[cite: 26, 28]. However, the unit sales forecast is in direct conflict: original reports confirmed Huang expects NVIDIA to sell twice as many chips next year, whereas a contradictory report claims **Huang expects NVIDIA to sell three times as many chips next year** (Article 109 ; $\Delta F_{contradiction} = 1.0$)[cite: 27, 28].

✓ Update Preserved: Samsung (+3.37%) & SK hynix (+6.42%) Rally on HBM4 Shift
✓ Conflict Preserved: Expects 2x Sales Volume (106/107) vs 3x (109)
✓ Duplicate Removed: Article 107 Consolidated

Group 044 — Standalone Summary

1 Article (110)

NVIDIA debuted preview MLPerf Inference v6.1 benchmark results for Vera Rubin NVL72 , demonstrating up to 3.7x higher throughput on Qwen3-VL and 2.5x higher throughput on DeepSeek-R1 compared to GB300 NVL72 (Article 110)[cite: 23, 28].

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 043

Batch 022 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

test_106_original.json — Group 043

```
{
  "id": 61862259,
  "headline": "Nvidia CEO Jensen Huang Expects Chip Sa
  "symbols": ["GOOG", "NVDA"],
  "summary": "Nvidia CEO Jensen Huang expects chip sal
}
```

Establishes 2x chip sales growth baseline.

test_108_update.json — Group 043

```
{
  "id": 990020008,
  "headline": "Nvidia CEO Vows to Double Chip Sales, L
  "symbols": ["NVDA"],
  "summary": "Nvidia CEO Jensen Huang said the company
}
```

Adds Samsung & SK hynix stock rally and HBM4 shift.

test_109_conflict.json — Group 043

```
{
  "id": 990020009,
  "headline": "Nvidia CEO Jensen Huang Expects Chip Sa
  "symbols": ["GOOG", "NVDA"],
  "summary": "Nvidia CEO Jensen Huang expects chip sal
}
```

Disputes 2x vs 3x sales growth guidance.